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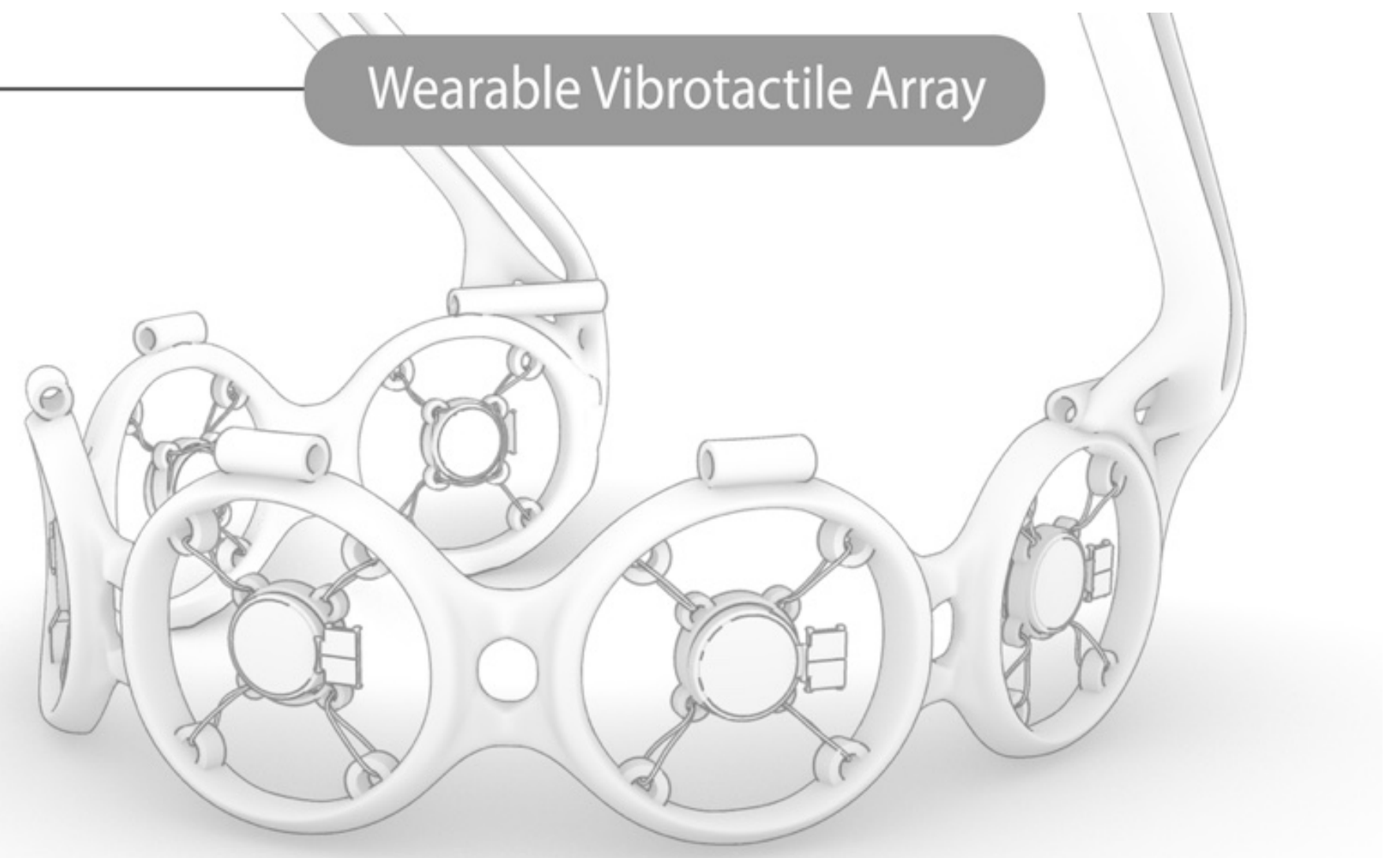
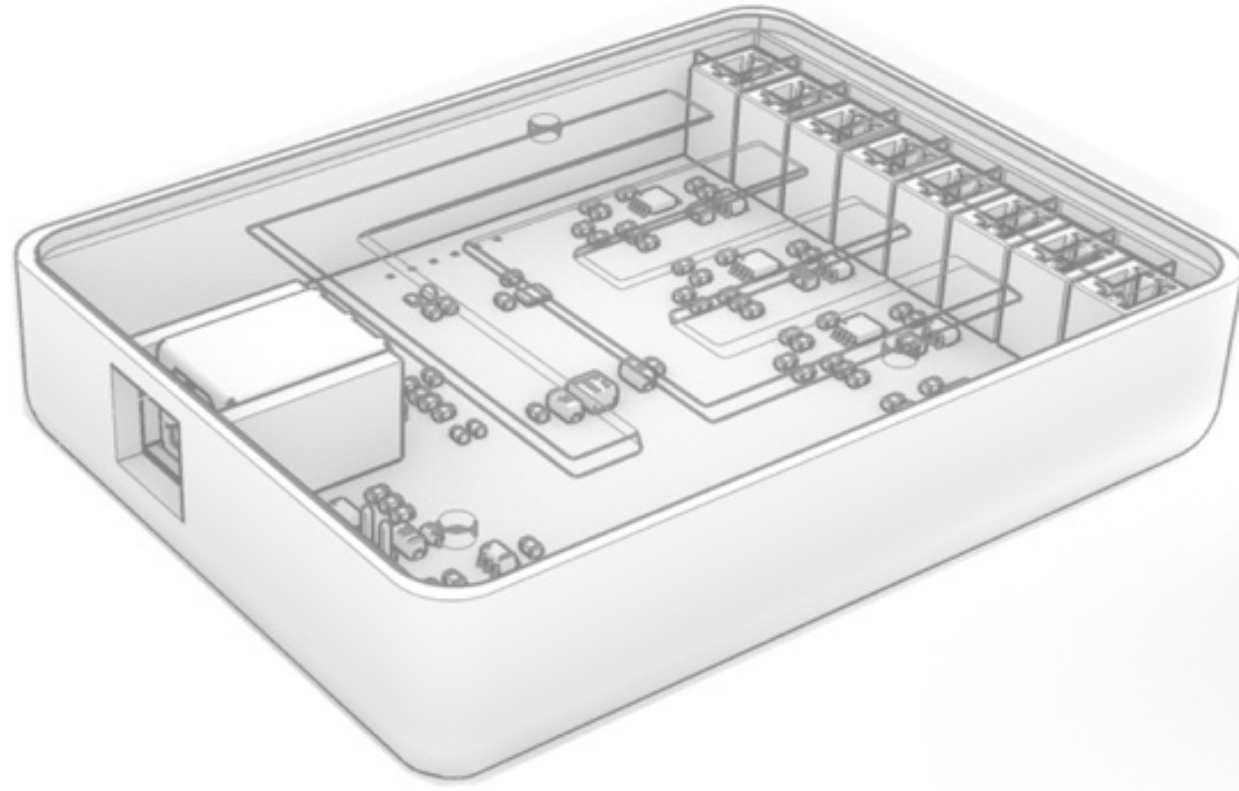
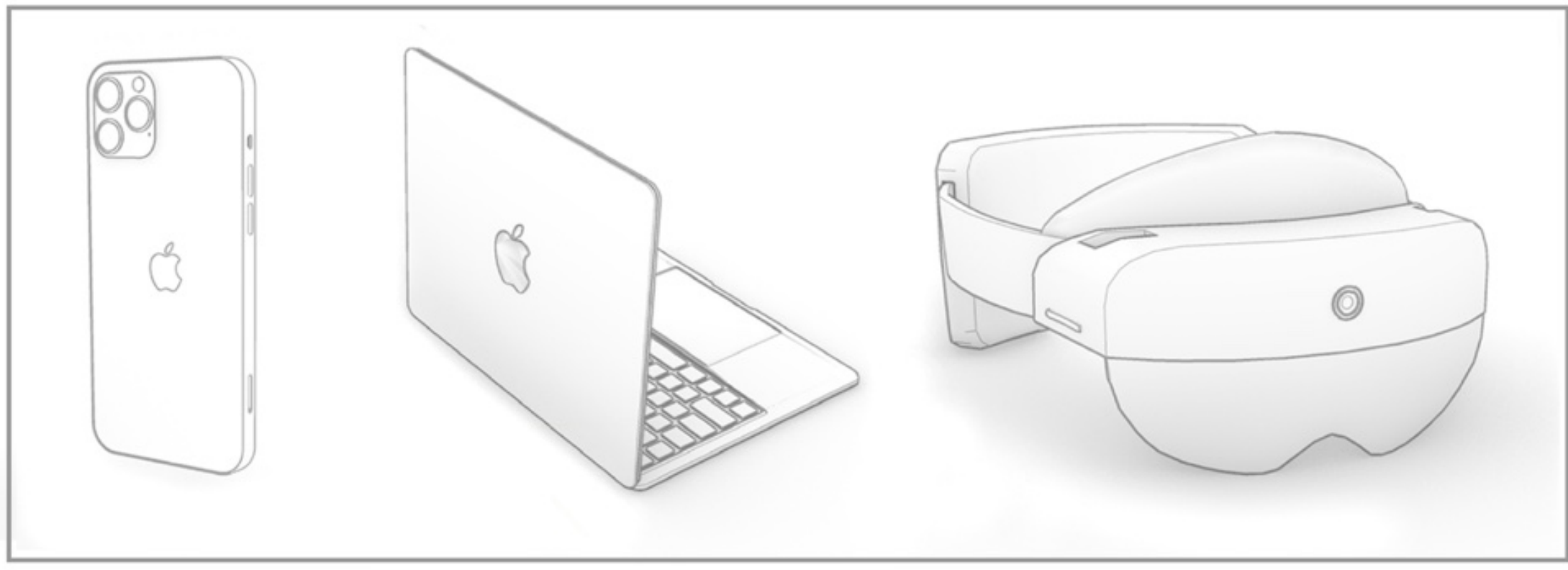
Demonstrating Sound2Haptic:

A Toolkit for Multichannel Haptics Across Multiple Form Factors

8 channel audio source

Haptic driver board

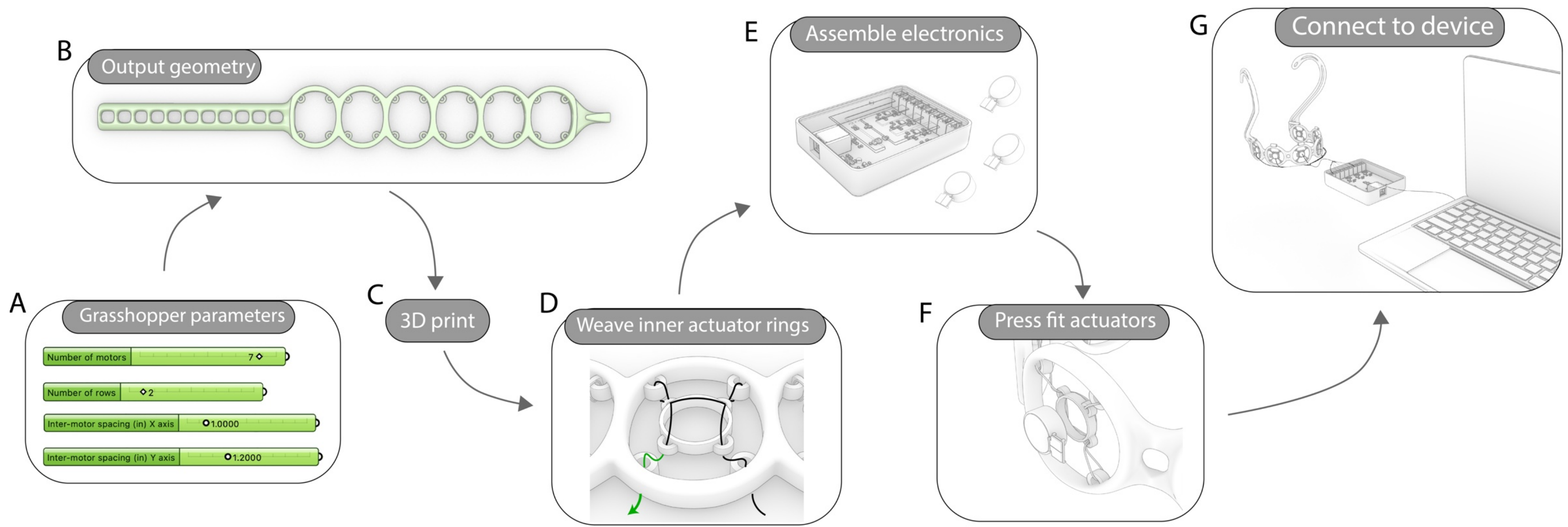
Wearable Vibrotactile Array



OVERVIEW

Sound2Haptic is a toolkit that solves two key problems in haptic prototyping. It replaces complex external hardware with a unified platform and presents a novel fabrication approach to create customizable devices.

FABRICATION PROCESS

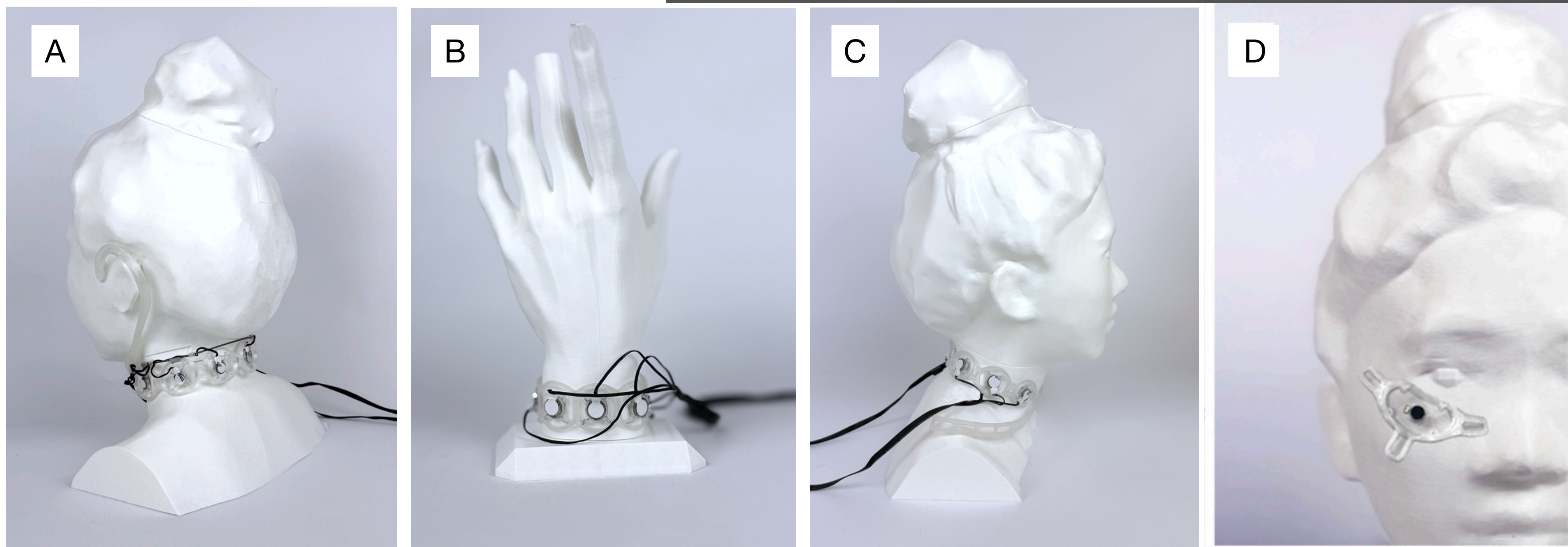


- (A) Sliders in Grasshopper control the array parameters.
(B) Grasshopper outputs 3D-printable geometry in Rhino.
(C) 3D-print the actuator array frame and inner actuator rings.
(D) Attach inner actuator rings to the frame with weaving process.
(E) Solder LRA actuators to wires, which connect to the PCB.
(F) Press fit actuators into the inner actuator rings.
(G) Plug the board into a phone, laptop, or XR headset via USB.

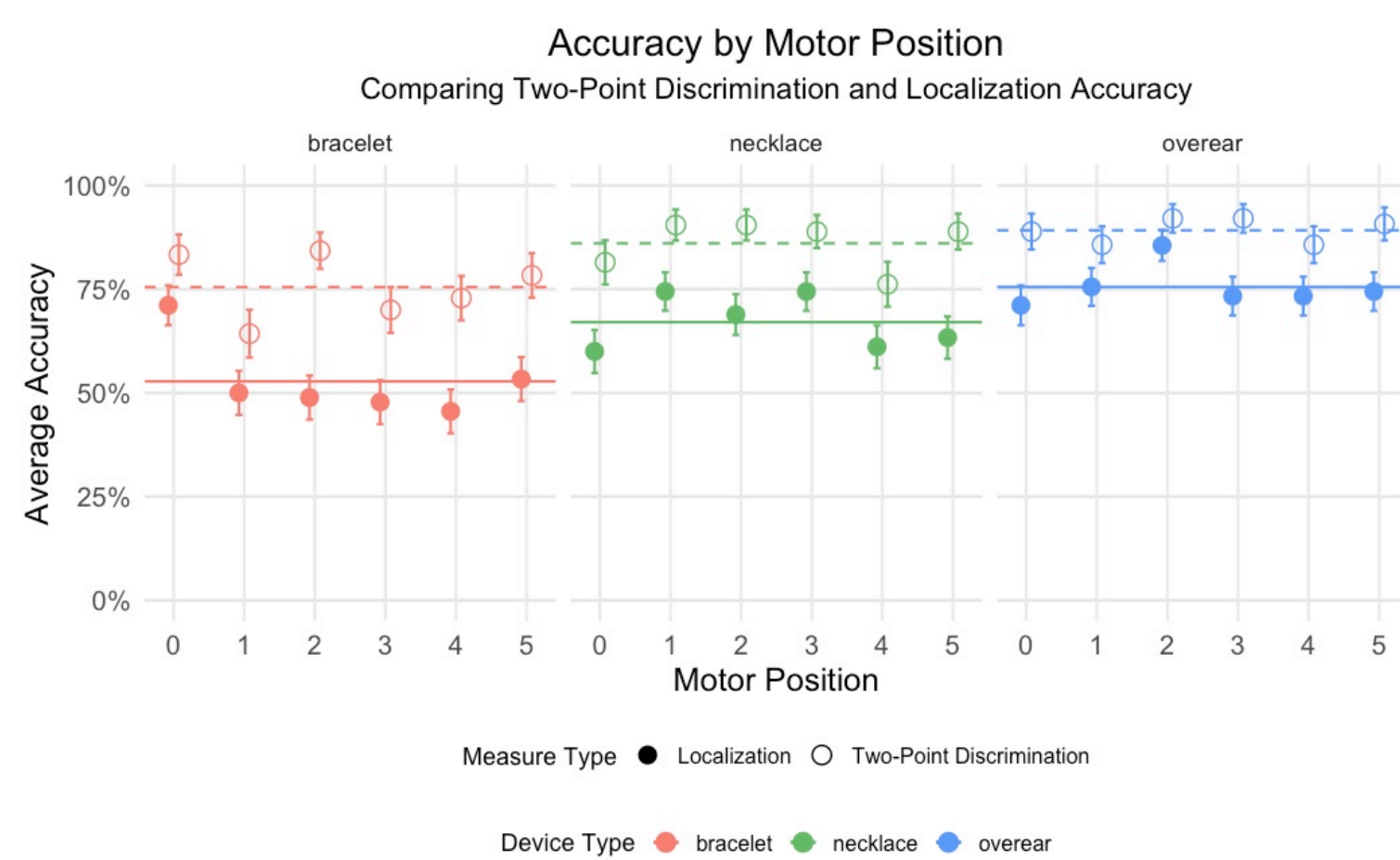
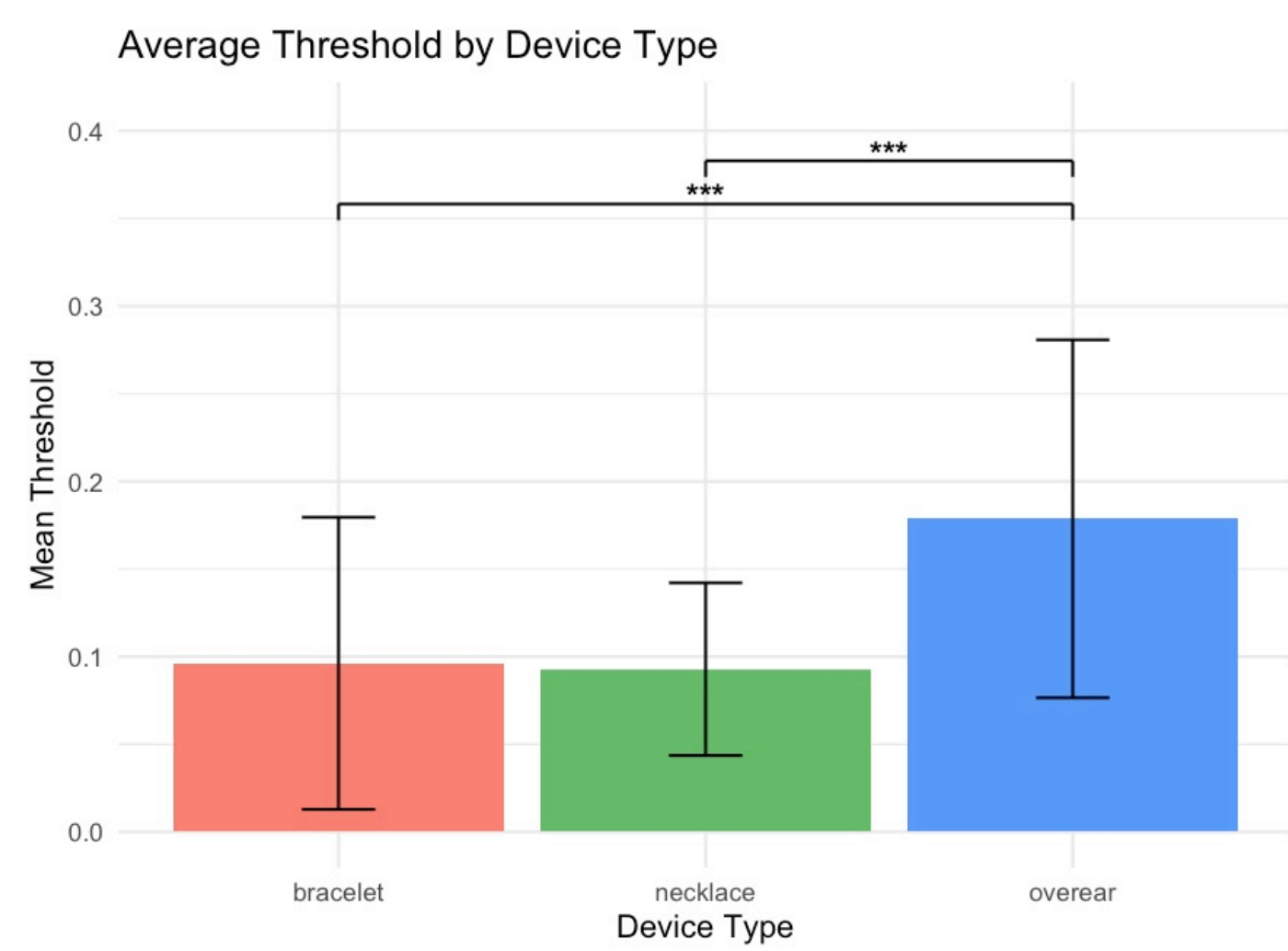
DEVICE SUMMARY

- **Plug-and-play device with 8-channels** of audio-based haptics. Combined sound card and amplifiers in one single device.
- 3D-printed design with a novel method of **passive vibration isolation** using actuators suspended with woven thread
- **Parametric design tool** in Grasshopper that allows users to easily define actuator layouts (e.g., number, spacing) and automatically generate 3D models for fabrication.

EXAMPLE ARTIFACTS



PSYCHOPHYSICAL EVALUATION



ELECTRONICS BLOCK DIAGRAM

